

A Signed Max-Cut Obstruction for Multi-Prime Cancellation

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Abstract

The preceding pairwise study found exact signed cancellation directions for individual physical prime components. This paper asks whether a whole prime shell admits one compatible coefficient-sign assignment. We formulate the question as a signed complete-graph max-cut problem: an edge is favorable when its signed Gram cross term becomes nonpositive. We prove that the all-positive K_m benchmark is $\lfloor m^2/4 \rfloor$, that the associated frustration index is exactly the complement of the signed maximum, and that vertex switching preserves the objective. An exact-rational replay of the inherited 18-row literal grid evaluates 1,380 shell edges and 5,727 triangles. Seventeen rows are all-positive and attain the benchmark; one early exponent-crossover row has three negative edges and a finite sign-only gain of three favorable edges. The result is a structural obstruction and a route selection device, not a magnitude-weighted energy estimate, an arithmetic L^2 theorem, or a twin-prime result.

1 Position on the twin-prime route

This release remains in the same literal prime-shell dynamical family. The earlier releases separated physical prime components and measured their Gram coherence [4, 3, 2]. Nonnegative shell weighting exposed a coherence wall [1], while TPC-291 showed that a single pair can be improved by an exact signed Schur direction [5]. TPC-292 then showed that three pair preferences obey a cycle-parity obstruction [6]. The smallest natural next question is global: does the entire signed shell have a useful sign assignment, or is the triangle obstruction only a local artifact?

We answer this question at the sign layer. This distinction matters. The physical norm contains the magnitudes of all Gram entries, whereas the graph objective below gives every nonzero edge unit weight. A sign-only gain is therefore a diagnostic and a clue for the next weighted experiment, not a claim of physical norm reduction.

2 Physical Gram shell

Let $I = I_N$ be the frozen integer interval and let β be the frozen rational source vector. For an odd prime q , height H , and exponent s , write

$$K_{H,s}(h) = \frac{H^{2s}}{(H^2 + h^2)^s}, \quad B_q(u, t) = \mathbf{1}_{q \nmid u} \mathbf{1}_{q \nmid t} \left(\mathbf{1}_{u \equiv t \pmod{q}} - \frac{1}{q-1} \right).$$

The deleted-diagonal physical component is $g_q = A_q \beta$, where

$$(A_q)_{u,t} = q K_{H,s}(u-t) \mathbf{1}_{u \neq t} B_q(u, t).$$

For a shell $S = \{q : Q < q \leq 2Q\}$, define

$$d_q = \|g_q\|^2, \quad G_{q,r} = \langle g_q, g_r \rangle, \quad \text{quad} \sigma_{q,r} = \text{sign}(G_{q,r}).$$

All finite calculations use exact rational arithmetic, so no floating-point rounding is used to decide an edge sign.

3 Signed-shell formulation

For vertex signs $a_q \in \{-1, +1\}$, call an edge favorable when

$$a_q a_r \sigma_{q,r} = -1.$$

The signed-shell value and frustration index are

$$F(\sigma) = \max_a \# \{(q, r) : a_q a_r \sigma_{q,r} = -1\}, \text{quad fr}(\sigma) = \binom{|S|}{2} - F(\sigma).$$

Fixing one label to +1 removes the global reversal symmetry without losing an optimum.

Theorem 1 (All-positive max-cut benchmark). *If $\sigma_{q,r} = +1$ on every edge of K_m , then*

$$F(\sigma) = \left\lfloor \frac{m^2}{4} \right\rfloor.$$

Proof. Let r vertices have label +1. A favorable edge is exactly an edge crossing the induced partition, so its number is $r(m-r)$. The identity

$$r(m-r) = \frac{m^2}{4} - \left(r - \frac{m}{2}\right)^2$$

gives the upper bound $\lfloor m^2/4 \rfloor$, and a partition with $r = \lfloor m/2 \rfloor$ attains it. $\square$

Lemma 1 (Frustration complement and switching). *For any finite signed graph with E edges, the minimum number of unfavorable edges is $E - F(\sigma)$. If vertex signs t_q switch the graph to $\sigma'_{q,r} = t_q \sigma_{q,r} t_r$, then $F(\sigma') = F(\sigma)$.*

Proof. For a fixed labeling every edge is either favorable or unfavorable, hence the unfavorable count is E minus the favorable count. Taking the minimum and maximum gives the first assertion. For the second, map a_q to $a'_q = t_q a_q$. Then $a'_q a'_r \sigma'_{q,r} = a_q a_r \sigma_{q,r}$ on every edge, so the map is a bijection preserving the objective. $\square$

Remark 1. *The graph objective is deliberately unweighted. The actual quadratic form is*

$$\left\| \sum_q c_q g_q \right\|^2 = \sum_q c_q^2 d_q + 2 \sum_{q < r} c_q c_r G_{q,r},$$

so replacing $|G_{q,r}|$ by one and restricting c_q to signs can change the optimizer. This paper makes that gap explicit.

4 Exact finite protocol

The certificate inherits eight $s = 2$ growth rows, four $s = 1$ exponent crossover rows, and six $s = 2$ source-control rows. For each row, every prime in $Q < q \leq 2Q$ is evaluated using the frozen source and physical operator. The resulting Gram matrix is exact; every off-diagonal sign is then placed in a complete signed graph. The producer enumerates all 2^{m-1} labelings, records one optimum and its count, and computes the triangle parity census as a consistency check.

The independent checker imports only the frozen TPC-268 engine and computes the physical outputs in source-coordinate-first order. It reconstructs the complete row records rather than importing the TPC-293 producer. Parent code/result hashes and the engine hash are recorded in both checkers.

5 Finite atlas

The edge signs are all positive in 17 rows. The sole exception is $(N, H, Q, z, s) = (256, 38, 27, 5, 1)$, whose seven-prime shell is $\{29, 31, 37, 41, 43, 47, 53\}$. Its three negative edges are $(29, 53)$, $(31, 53)$, and $(41, 53)$. Consequently its signed maximum is 15, compared with the all-positive benchmark 12; its minimum unsatisfied count is 6 instead of 9. The aggregate gain is therefore exactly 3 edges.

The 5,727 triangles contain 5,718 sign-frustrated triangles, consistent with the TPC-292 parity census. In particular, the late row $(512, 58, 90, 5, 2)$ contributes all 680 possible triangles with ‘+++’ edge pattern. The shell-level optimization therefore does not turn the late near-coherent block into a uniformly favorable cancellation direction.

Table 1: Global exact-rational signed-shell census.

quantity	count
rows	18
shell edges	1,380
maximum favorable edges (summed)	744
minimum unsatisfied edges (summed)	636
signed gain over all-positive benchmark	3
all-positive rows	17
signed-gain rows	1
triangles	5,727
sign-frustrated triangles	5,718

Table 2: Representative row records. The frustration column is the exact minimum number of unsatisfied sign edges.

axis	N	H	Q	s	$ S $	F	fr	gain
growth	128	24	9	2	3	2	1	0
growth	192	32	16	2	5	6	4	0
growth	256	38	27	2	7	12	9	0
growth	384	50	40	2	10	25	20	0
growth	512	58	60	2	13	42	36	0
growth	512	58	90	2	17	72	64	0
crossover	256	38	27	1	7	15	6	3
control	384	52	70	2	15	56	49	0

6 Interpretation and claim firewall

The result closes one structural uncertainty and leaves the arithmetic road unchanged. On the declared grid, the signed graph is overwhelmingly the all-positive graph, for which the exact max-cut benchmark is already sharp. The exceptional crossover row is useful precisely because it is small: it shows that sign irregularity can exist, but it supplies only a unit-weight gain and no evidence that Gram magnitudes or source coefficients exploit it.

The exact claims are the max-cut theorem, the frustration complement, and switching invariance. The finite atlas is certified by exact rational reproduction and an independent physical replay. A growing-shell signed graph theorem, a magnitude-weighted Rayleigh estimate, a source-restricted coefficient-image theorem, an arithmetic L^2 bound, fixed-power credit, and the full Gate-B/twin-prime endpoint remain open. The Session-named Route-A and Route-B evaluator files are absent from this checkout; no official evaluator pass is claimed.

7 Reproducibility and conclusion

From the project directory, run:

```
export PYTHONDONTWRITEBYTECODE=1
python -B code/tpc293_signed_shell_maxcut_certificate.py --check
python -B experiments/tpc293_independent_checker.py
python -B experiments/tpc293_signed_graph_stress.py
```

The canonical finite result is `results/tpc293_certificate.json`. Normal and optimized runs are required to have empty standard error and byte-identical standard output.

TPC-293 promotes the route’s reusable object from a signed triangle to a signed prime-shell graph. The next logically minimal experiment is to put the actual Gram magnitudes back into the objective and test whether the finite ‘+3’ sign anomaly survives.

References

- [1] Liang Wang. Adaptive shell weighting and the nonnegative coherence wall, 2026. TPC-290 project release in the same research repository.
- [2] Liang Wang. Cross-prime gram coherence in a literal twin-prime dynamics, 2026. TPC-289 project release in the same research repository.

- [3] Liang Wang. Growing-shell gram obstruction in a literal twin-prime dynamics, 2026. TPC-288 project release in the same research repository.
- [4] Liang Wang. Prime-shell cancellation depth in a literal twin-prime dynamics, 2026. TPC-287 project release in the same research repository.
- [5] Liang Wang. Signed schur cancellation in cross-prime gram pairs, 2026. TPC-291 project release in the same research repository.
- [6] Liang Wang. Three-prime sign frustration in a physical gram shell, 2026. TPC-292 project release in the same research repository.